

Histopathological Image Classification of Gastric Cancer Related Tissues Using CNN Feature Extraction and Traditional Machine Learning - Project Proposal

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Abstract—Gastric cancer, also known as stomach cancer, is the drastic growth of abnormal cells and tissues within the stomach, which would form tumors.

Gastric cancer is one of the leading causes of cancer death with approximately 11000 people in the US dying of gastric cancer each year. Because of this, accurate early diagnosis is required to improve survival rates. Traditional diagnosis based on histopathological image analysis is time-consuming and subjective, heavily relying on skilled pathologists. The project addresses the challenge of automating gastric cancer detection from histopathology image patches by developing a hybrid classification pipeline.

Keywords—Gastric Cancer, Machine Learning, CNN, Image Analysis

1. Project description

This project will be utilizing a pretrained convolutional neural network (ResNet50) to extract high-level visual features from tissue images, and traditional machine learning models—e.g., Support Vector Machines and Random Forests—are used for the final classification. By separating deep feature extraction from the classification task, the approach is consistent with limitations on the use of end-to-end deep learning. With over 31,000 labeled gastric tissue images from a real-world dataset on Kaggle, the project aims to create an accurate, interpretable, and scalable classification system to support clinical diagnostic procedures

2. Dataset

Gastric Cancer Histopathology Tissue Image Dataset taken from Kaggle (the HMU-GC-HE-30K subset, specifically) contains histopathological images of gastric tissue under H&E staining. It is divided into 8 various classes, each representing various tissue types for gastric cancer diagnosis:

- ADI - Adipose tissue
- DEB - Debris
- LYM - Lymphocytes
- MUC - Mucus
- MUS - Smooth Muscle
- NORM - Normal Colon Mucosa
- STR - Cancer-associated Stroma - Cancerous
- TUM - Tumor - Cancerous

Each class contains 3887 images, totaling of 31096 images combined. The images are in RGB and vary in size with dimensions of 224x224.

2.1. Data Exploration and Analysis

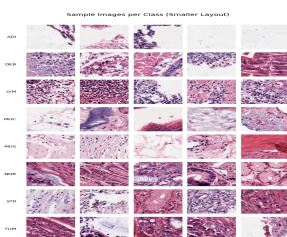


Figure 1. Sample Images for each Class

Figure 1 shows a stratified sample of five randomly selected images from each class was extracted and visualized to provide a representative overview of the dataset's visual characteristics.

The visualizations also suggest that the color features will be key to the classification problem since they correspond to biologically informative structures—e.g., purplish nuclei and cytoplasmic or stromal regions pink-stained by H&E. These color features should prove helpful in enabling the model to differentiate between tissue types.

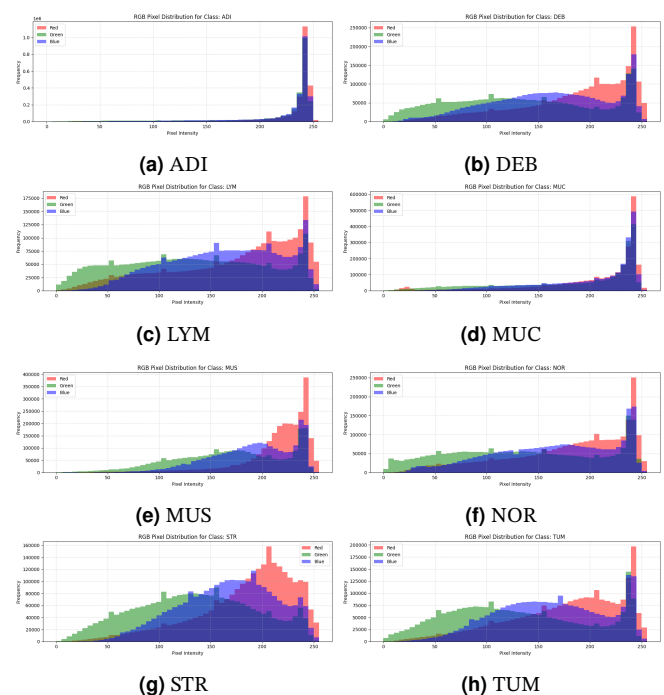


Figure 2. Channel Distribution across Classes

The 8 figures presented above show us the distribution of color channels (RGB) across the 8 different classes. In H&E-stained histopathology images, each color channel carries biological significance. The blue channel represents hematoxylin staining, which will bind to cell nuclei; therefore, a high frequency of blue pixel might indicate dense cellular concentration. The red channel reflects the eosin staining, which suggests the presence of cytoplasm, connective tissue, and mucin. Together, these colors provide great insight and essential visual cues for distinguishing between different tissues. However, the color information alone is not adequate for accurate classification because there is overlapping chromatic distribution of some classes—such as ADI and MUC. Thus, spatial and textural features must be extracted, and these can be effectively obtained by employing Convolutional Neural Network (CNN) architectures.

3. Methodology

The HMU-GC-HE-30K dataset primarily consists of gastric histopathology image patches categorized into eight distinct tissue types: ADI, MUS, NOR, LYM, MUC, STR, DEB, and TUM. However,

it is important to note that the dataset does not provide explicit labels indicating whether a given image represents a health threat (e.g., cancerous or precancerous). With the exception of STR (stroma) and TUM (tumor epithelium), which are more strongly associated with malignancy, the remaining tissue classes can represent both normal physiological components and benign or reactive changes. As a result, the existing class labels alone are insufficient for direct cancer detection. To address this limitation, an additional step must be incorporated into the classification pipeline. Specifically, after initially classifying each image by tissue type, a secondary process—such as class-specific anomaly detection or unsupervised outlier detection—will be applied to identify deviations from the expected appearance of healthy tissue within that class.

General pipeline:

1. **Feature Extraction:** Use a pre-trained ResNet-50 model to extract features from each image without training the network.
2. **Tissue Classification:** Train a traditional machine learning model to classify images into one of the eight tissue types.
3. **Unsupervised Anomaly Detection:** For each tissue class, apply an anomaly detection method (e.g., Isolation Forest, One-Class SVM) to pseudo-label images as 'normal' or 'abnormal'.
4. **Abnormality Classification:** Train a class-specific model to distinguish normal from suspicious images using the pseudo-labels.
5. **Flagging Suspicious Regions:** During inference, classify the image by tissue type, assess abnormality using the corresponding model, and flag it if suspicious.

3.1. Feature Extraction

In this project, ResNet-50 is used as a pre-trained feature extractor. It converts each histopathology image into a 2048-dimensional feature vector, capturing important visual patterns. These features are then passed to traditional machine learning models for tasks like tissue classification and abnormality detection, without training the CNN itself.

```
1 model = ResNet50(weights='imagenet', include_top=False,
2   ↪ pooling='avg')
3
4 def extract_features(img_path):
5     img = image.load_img(img_path, target_size=(224, 224
6     ↪ ))
7     x = image.img_to_array(img)
8     x = np.expand_dims(x, axis=0)
9     x = preprocess_input(x)
10    features = model.predict(x, verbose=0)
11    return features.flatten()
```

Code 1. Feature extraction using ResNet-50

3.2. Tissue Classification

3.2.1. Data Preprocessing

After feature extraction, the data undergoes a preprocessing phase that includes dimensionality reduction techniques such as Principal Component Analysis (PCA) or Singular Value Decomposition (SVD). They transform the high-dimensional feature space into a low-dimensional subspace in a way that retains most of the top variance in the data. The step has the effect of removing noise, removing redundant or highly correlated features, and reducing computational complexity, thereby improving the efficiency and generalization of the classification and anomaly detection models that follow.

3.2.2. Classification Model

We will be using traditional machine learning models in order to train the dataset. The possible methods that we are going to use are Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors

(K-NN) and Logistic Regressions. These models are chosen because they are able to work well with high dimensional and non-linear data.

3.3. Anomaly Detection - Unsupervised Training

The second model that is going to be integrated into the proposed pipeline is a class-specific anomaly detection model, designed to identify abnormal tissue appearances within each individual tissue class.

This process is conducted using traditional unsupervised machine learning methods such as:

1. Isolation Forest
2. One-Class SVM
3. Clustering-Based

3.3.1. Alternatives

Another approach is to utilize the STR and TUM classes as training samples of abnormal tissue patterns. Both being highly related to malignant histological appearance, they would be ideal samples of what "abnormal" would look like. A one-class anomaly detection model can be trained on feature vectors extracted from STR and TUM patches alone. The model will learn the pattern of abnormal features without examples of normal tissue. At inference, images belonging to other classes of tissues—such as ADI, MUS, or NOR—can be input through this model to test whether they resemble the abnormal reference set. A patch can be labeled as suspicious if it matches the learned abnormal pattern closely. This method allows cross-class abnormality detection without the need for labeling normal data, making it an unsupervised and scalable way to detect potentially malignant regions.

4. References

References

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